THE DETERMINANT OF BIT MATRICES (OPTIONAL)

The properties and techniques for the determinant developed in this section apply to bit matrices, where computations are carried out using binary arithmetic.

EXAMPLE 20 The determinant of the 2×2 bit matrix

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

computed by using the technique developed in Example 5 is

$$\det(A) = (1)(1) - (1)(0) = 1.$$

■

EXAMPLE 21 The determinant of the 3×3 bit matrix

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

computed by using the technique developed in Example 6 is

$$\begin{aligned} \det(A) &= (1)(1)(1) + (0)(0)(0) + (1)(1)(1) \\ &\quad - (1)(0)(1) - (1)(0)(1) - (0)(1)(1) \\ &= 1 + 0 + 1 - 0 - 0 - 0 = 1 + 1 = 0. \end{aligned}$$

■

EXAMPLE 22 Use the computation via reduction to triangular form to evaluate the determinant of the bit matrix

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}.$$

Solution

$$\begin{vmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{vmatrix}_{\mathbf{r}_1 \leftrightarrow \mathbf{r}_2} = (-1) \begin{vmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{vmatrix}_{\mathbf{r}_1 + \mathbf{r}_3 \to \mathbf{r}_3} = (-1) \begin{vmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{vmatrix}$$

By Theorem 3.3, $\det(A) = 0$. ■

Key Terms

Permutation
n factorial
Inversion
Even permutation
Odd permutation
Determinant
Computation via reduction to triangular form

3.1 Exercises

1. Find the number of inversions in each of the following permutations of $S = \{1, 2, 3, 4, 5\}$.

   (a) 52134 (b) 45213 (c) 42135

   (d) 13542 (e) 35241 (f) 12345

2. Determine whether each of the following permutations of $S = \{1, 2, 3, 4\}$ is even or odd.

   (a) 4213 (b) 1243 (c) 1234

   (d) 3214 (e) 1423 (f) 2431

3. Determine the sign associated with each of the following permutations of $S = \{1, 2, 3, 4, 5\}$.

(a) 25431 (b) 31245 (c) 21345

(d) 52341 (e) 34125 (f) 41253

4. In each of the following pairs of permutations of $S = \{1, 2, 3, 4, 5, 6\}$, verify that the number of inversions differs by an odd number.

(a) 436215 and 416235

(b) 623415 and 523416

(c) 321564 and 341562

(d) 123564 and 423561

In Exercises 5 and 6, evaluate the determinants using Equation (2).

5. (a) $\begin{vmatrix} 2 & -1 \\ 3 & 2 \end{vmatrix}$ (b) $\begin{vmatrix} 0 & 3 & 0 \\ 2 & 0 & 0 \\ 0 & 0 & -5 \end{vmatrix}$

(c) $\begin{vmatrix} 4 & 2 & 0 \\ 0 & -2 & 5 \\ 0 & 0 & 3 \end{vmatrix}$ (d) $\begin{vmatrix} 4 & 2 & 2 & 0 \\ 2 & 0 & 0 & 0 \\ 3 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{vmatrix}$

6. (a) $\begin{vmatrix} 2 & 1 \\ 4 & 3 \end{vmatrix}$ (b) $\begin{vmatrix} 0 & 0 & -2 \\ 0 & 3 & 0 \\ 4 & 0 & 0 \end{vmatrix}$

(c) $\begin{vmatrix} 3 & 4 & 2 \\ 2 & 5 & 0 \\ 3 & 0 & 0 \end{vmatrix}$ (d) $\begin{vmatrix} -4 & 2 & 0 & 0 \\ 2 & 3 & 1 & 0 \\ 3 & 1 & 0 & 2 \\ 1 & 3 & 0 & 3 \end{vmatrix}$

7. Let $A = \left[a_{ij}\right]$ be a 4×4 matrix. Write the general expression for $\det(A)$ using Equation (2).

8. If

$$|A| = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = -4,$$

find the determinants of the following matrices:

$$B = \begin{bmatrix} a_3 & a_2 & a_1 \\ b_3 & b_2 & b_1 \\ c_3 & c_2 & c_1 \end{bmatrix},$$

$$C = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ 2c_1 & 2c_2 & 2c_3 \end{bmatrix},$$

and

$$D = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 + 4c_1 & b_2 + 4c_2 & b_3 + 4c_3 \\ c_1 & c_2 & c_3 \end{bmatrix}.$$

9. If

$$|A| = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = 3,$$

find the determinants of the following matrices:

$B =$

$$\begin{bmatrix} a_1 + 2b_1 - 3c_1 & a_2 + 2b_2 - 3c_2 & a_3 + 2b_3 - 3c_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix},$$

$$C = \begin{bmatrix} a_1 & 3a_2 & a_3 \\ b_1 & 3b_2 & b_3 \\ c_1 & 3c_2 & c_3 \end{bmatrix},$$

and

$$D = \begin{bmatrix} a_1 & a_2 & a_3 \\ c_1 & c_2 & c_3 \\ b_1 & b_2 & b_3 \end{bmatrix}.$$

10. If

$$A = \begin{bmatrix} 1 & -1 & 2 \\ 3 & 4 & 1 \\ 2 & 5 & 1 \end{bmatrix},$$

verify that $\det(A) = \det(A^T)$.

11. Evaluate:

(a) $\det\left(\begin{bmatrix} \lambda - 1 & 2 \\ 3 & \lambda - 2 \end{bmatrix}\right)$

(b) $\det(\lambda I_2 - A)$, where $A = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$

12. Evaluate:

(a) $\det\left(\begin{bmatrix} \lambda - 1 & -1 & -2 \\ 0 & \lambda - 2 & 2 \\ 0 & 0 & \lambda - 3 \end{bmatrix}\right)$

(b) $\det(\lambda I_3 - A)$, where $A = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & -1 \\ 0 & 0 & 1 \end{bmatrix}$

13. For each of the matrices in Exercise 11, find all values of λ for which the determinant is 0.

14. For each of the matrices in Exercise 12, find all values of λ for which the determinant is 0.

In Exercises 15 and 16, compute the indicated determinant.

15. (a) $\begin{vmatrix} 0 & 2 & -5 \\ 0 & 4 & 6 \\ 0 & 0 & -1 \end{vmatrix}$ (b) $\begin{vmatrix} 6 & 6 & 3 & -2 \\ 0 & 4 & 7 & 5 \\ 0 & 0 & -3 & 2 \\ 0 & 0 & 0 & 2 \end{vmatrix}$

(c) $\begin{vmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 9 \end{vmatrix}$

16. (a) $\begin{vmatrix} 6 & 0 & 0 & 0 \\ 3 & 4 & 0 & 0 \\ 5 & 9 & -3 & 0 \\ 4 & 1 & -3 & 2 \end{vmatrix}$ (b) $\begin{vmatrix} 7 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & -3 \end{vmatrix}$

(c) $\begin{vmatrix} 2 & 6 & -5 \\ 0 & 4 & 0 \\ 0 & 0 & 9 \end{vmatrix}$

In Exercises 17 through 20, evaluate the given determinant via reduction to triangular form.

17. (a) $\begin{vmatrix} 4 & -3 & 5 \\ 5 & 2 & 0 \\ 2 & 0 & 4 \end{vmatrix}$ (b) $\begin{vmatrix} 2 & 0 & 1 & 4 \\ 3 & 2 & -4 & -2 \\ 2 & 3 & -1 & 0 \\ 11 & 8 & -4 & 6 \end{vmatrix}$

(c) $\begin{vmatrix} 4 & 1 & 2 \\ 0 & 2 & 3 \\ 0 & 0 & -3 \end{vmatrix}$

18. (a) $\begin{vmatrix} 4 & 0 & 0 & 0 \\ -1 & 2 & 0 & 0 \\ 1 & 2 & -3 & 0 \\ 1 & 5 & 3 & 5 \end{vmatrix}$ (b) $\begin{vmatrix} 4 & 1 & 3 \\ 2 & 3 & 0 \\ 1 & 3 & 2 \end{vmatrix}$

(c) $\begin{vmatrix} 1 & 2 & 3 \\ 2 & 1 & 0 \\ -3 & 1 & 2 \end{vmatrix}$

19. (a) $\begin{vmatrix} 4 & 2 & 3 & -4 \\ 3 & -2 & 1 & 5 \\ -2 & 0 & 1 & -3 \\ 8 & -2 & 6 & 4 \end{vmatrix}$

(b) $\begin{vmatrix} 1 & 3 & -4 \\ -2 & 1 & 2 \\ -9 & 15 & 0 \end{vmatrix}$ (c) $\begin{vmatrix} 1 & 1 & 2 \\ 0 & 2 & -2 \\ 0 & 0 & 3 \end{vmatrix}$

20. (a) $\begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 2 & 1 & 0 \end{vmatrix}$ (b) $\begin{vmatrix} 2 & 0 & 0 & 0 \\ -5 & 3 & 0 & 0 \\ 3 & 2 & 4 & 0 \\ 4 & 2 & 1 & -5 \end{vmatrix}$

(c) $\begin{vmatrix} 1 & 2 & -1 \\ 3 & 2 & 0 \\ 1 & 4 & 3 \end{vmatrix}$

21. Verify that $\det(AB) = \det(A)\det(B)$ for the following:

(a) $A = \begin{bmatrix} 1 & -2 & 3 \\ -2 & 3 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 & 2 \\ 3 & -2 & 5 \\ 2 & 1 & 3 \end{bmatrix}$

(b) $A = \begin{bmatrix} 2 & 3 & 6 \\ 0 & 3 & 2 \\ 0 & 0 & -4 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 0 & 0 \\ 4 & 5 & 0 \\ 2 & 1 & -2 \end{bmatrix}$

22. If $|A| = -4$, find

(a) $|A^2|$ (b) $|A^4|$ (c) $|A^{-1}|$

23. If A and B are $n \times n$ matrices with $|A| = 2$ and $|B| = -3$, calculate $|A^{-1}B^T|$.

In Exercises 24 and 25, evaluate the given determinant of the bit matrices using techniques developed in Examples 5 and 6.

24. (a) $\begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix}$ (b) $\begin{vmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{vmatrix}$

(c) $\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{vmatrix}$

25. (a) $\begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix}$ (b) $\begin{vmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{vmatrix}$

(c) $\begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \end{vmatrix}$

In Exercises 26 and 27, evaluate the given determinant of the bit matrices via reduction to triangular form.

26. (a) $\begin{vmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{vmatrix}$ (b) $\begin{vmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{vmatrix}$

27. (a) $\begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{vmatrix}$ (b) $\begin{vmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{vmatrix}$

Theoretical Exercises

T.1. Show that if we interchange two numbers in the permutation $j_1 j_2 \cdots j_n$, then the number of inversions is either increased or decreased by an odd number. (*Hint*: First show that if two adjacent numbers are interchanged, the number of inversions is either increased or decreased by 1. Then show that an interchange of any two numbers can be achieved by an odd number of successive interchanges of adjacent numbers.)

T.2. Prove Theorem 3.7 for the lower triangular case.

T.3. Show that if c is a real number and A is $n \times n$, then $\det(cA) = c^n \det(A)$.

T.4. Prove Corollary 3.2.

T.5. Show that if $\det(AB) = 0$, then $\det(A) = 0$ or $\det(B) = 0$.

T.6. Is $\det(AB) = \det(BA)$? Justify your answer.

T.7. Show that if A is a matrix such that in each row and in each column one and only one element is $\neq 0$, then $\det(A) \neq 0$.

T.8. Show that if $AB = I_n$, then $\det(A) \neq 0$ and $\det(B) \neq 0$.

T.9. (a) Show that if $A = A^{-1}$, then $\det(A) = \pm 1$.

(b) Show that if $A^T = A^{-1}$, then $\det(A) = \pm 1$.

T.10. Show that if A is a nonsingular matrix such that $A^2 = A$, then $\det(A) = 1$.

T.11. Show that

$$\det(A^T B^T) = \det(A)\det(B^T) = \det(A^T)\det(B).$$

T.12. Show that

$$\begin{vmatrix} a^2 & a & 1 \\ b^2 & b & 1 \\ c^2 & c & 1 \end{vmatrix} = (b-a)(c-a)(b-c).$$

This determinant is called a **Vandermonde*** **determinant.**

T.13. Let $A = \left[a_{ij}\right]$ be an upper triangular matrix. Show that A is nonsingular if and only if $a_{ii} \neq 0$, $1 \leq i \leq n$.

T.14. Show that if $\det(A) = 0$, then $\det(AB) = 0$.

T.15. Show that if $A^n = O$, for some positive integer n, then $\det(A) = 0$.

T.16. Show that if A is $n \times n$, with A skew symmetric ($A^T = -A$, see Section 1.4, Exercise T.24), and n odd, then $\det(A) = 0$.

T.17. Prove Corollary 3.1.

T.18. When is a diagonal matrix nonsingular? (*Hint*: See Exercise T.7.)

T.19. Using Exercise T.13 in Section 1.2, determine how many 2×2 bit matrices have determinant 0 and how many have determinant 1.

MATLAB Exercises

In order to use MATLAB *in this section, you should first have read Chapter 12 through Section 12.5.*

ML.1. Use the routine **reduce** to perform row operations and keep track by hand of the changes in the determinant as in Example 17.

(a) $A = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 3 & 2 \\ 3 & 2 & 1 \end{bmatrix}$

(b) $A = \begin{bmatrix} 0 & 1 & 3 & -2 \\ -2 & 1 & 1 & 1 \\ 2 & 0 & 1 & 2 \\ 1 & 0 & 0 & 1 \end{bmatrix}$

ML.2. Use routine **reduce** to perform row operations and keep track by hand of the changes in the determinant as in Example 17.

(a) $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 1 & 0 \end{bmatrix}$

(b) $A = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 2 & 1 & 2 & 0 \\ 0 & 2 & 1 & 2 \\ 0 & 0 & 2 & 1 \end{bmatrix}$

ML.3. MATLAB has command **det**, which returns the value of the determinant of a matrix. Just type **det(A)**. Find the determinant of each of the following matrices using **det**.

(a) $A = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{bmatrix}$

(b) $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \\ 4 & 5 & 6 & 7 \end{bmatrix}$

ML.4. Use **det** (see Exercise ML.3) to compute the determinant of each of the following.

(a) **5 ∗ eye(size(A)) − A**, where

$$A = \begin{bmatrix} 2 & 3 & 0 \\ 4 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix}.$$

(b) **(3 ∗ eye(size(A)) − A)^2**, where

$$A = \begin{bmatrix} 1 & 1 \\ 5 & 2 \end{bmatrix}.$$

(c) **invert(A) ∗ A**, where

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}.$$

ML.5. Determine a positive integer t so that **det(t ∗ eye(size(A)) − A) = 0**, where

$$A = \begin{bmatrix} 5 & 2 \\ -1 & 2 \end{bmatrix}.$$

*Alexandre-Théophile Vandermonde (1735–1796) was born in Paris. His father, a physician, tried to steer him toward a musical career. His published mathematical work consisted of four papers that were presented over a two-year period. He is generally considered the founder of the theory of determinants and also developed formulas for solving general quadratic, cubic, and quartic equations. Vandermonde was a cofounder of the Conservatoire des Arts et Métiers and was its director from 1782. In 1795 he helped to develop a course in political economy. He was an active revolutionary in the French Revolution and was a member of the Commune of Paris and the club of the Jacobins.

POLYNOMIAL INTERPOLATION REVISITED

At the end of Section 1.7 we discussed the problem of finding a quadratic polynomial that interpolates the points (x_1, y_1), (x_2, y_2), (x_3, y_3), $x_1 \neq x_2$, $x_1 \neq x_3$, and $x_2 \neq x_3$. Thus, the polynomial has the form

$$y = a_2x^2 + a_1x + a_0 \tag{9}$$

[this was Equation (15) in Section 1.6]. Substituting the given points in (9), we obtain the linear system

$$\begin{aligned} a_2x_1^2 + a_1x_1 + a_0 &= y_1 \\ a_2x_2^2 + a_1x_2 + a_0 &= y_2 \\ a_2x_3^2 + a_1x_3 + a_0 &= y_3. \end{aligned} \tag{10}$$

The coefficient matrix of this linear system is

$$\begin{bmatrix} x_1^2 & x_1 & 1 \\ x_2^2 & x_2 & 1 \\ x_3^2 & x_3 & 1 \end{bmatrix}$$

whose determinant is the Vandermonde determinant (see Exercise T.12 in Section 3.1), which has the value

$$(x_2 - x_1)(x_3 - x_1)(x_2 - x_3).$$

Since the three given points are *distinct*, the Vandermonde determinant is not zero. Hence, the coefficient matrix of the linear system in (10) is nonsingular, which implies that the linear system has a unique solution. Thus there is a unique interpolating quadratic polynomial. The general proof for n points is similar.

OTHER APPLICATIONS OF DETERMINANTS

In Section 4.1 we use determinants to compute the area of a triangle and in Section 5.1 to compute the area of a parallelepiped.

Key Terms

Minor
Cofactor
Adjoint

3.2 Exercises

1. Let

$$A = \begin{bmatrix} 1 & 0 & -2 \\ 3 & 1 & 4 \\ 5 & 2 & -3 \end{bmatrix}.$$

Compute all the cofactors.

2. Let

$$A = \begin{bmatrix} 1 & 0 & 3 & 0 \\ 2 & 1 & 4 & -1 \\ 3 & 2 & 4 & 0 \\ 0 & 3 & -1 & 0 \end{bmatrix}.$$

Compute all the cofactors of the elements in the second row and all the cofactors of the elements in the third column.

In Exercises 3 through 6, evaluate the determinants using Theorem 3.9.

3. (a) $\begin{vmatrix} 1 & 2 & 3 \\ -1 & 5 & 2 \\ 3 & 2 & 0 \end{vmatrix}$

(b) $\begin{vmatrix} 4 & -4 & 2 & 1 \\ 1 & 2 & 0 & 3 \\ 2 & 0 & 3 & 4 \\ 0 & -3 & 2 & 1 \end{vmatrix}$

(c) $\begin{vmatrix} 4 & -2 & 0 \\ 0 & 2 & 4 \\ -1 & -1 & -3 \end{vmatrix}$

4. (a) $\begin{vmatrix} 2 & 2 & -3 & 1 \\ 0 & 1 & 2 & -1 \\ 3 & -1 & 4 & 1 \\ 2 & 3 & 0 & 0 \end{vmatrix}$

(b) $\begin{vmatrix} 0 & 1 & -2 \\ -1 & 3 & 1 \\ 2 & -2 & 3 \end{vmatrix}$

(c) $\begin{vmatrix} 2 & 1 & -3 \\ 0 & 1 & 2 \\ -4 & 2 & 1 \end{vmatrix}$

5. (a) $\begin{vmatrix} 3 & 1 & 2 & -1 \\ 2 & 0 & 3 & -7 \\ 1 & 3 & 4 & -5 \\ 0 & -1 & 1 & -5 \end{vmatrix}$

(b) $\begin{vmatrix} 3 & 1 & 0 \\ 3 & 2 & 1 \\ 0 & 1 & -1 \end{vmatrix}$

(c) $\begin{vmatrix} 3 & -3 & 0 \\ 2 & 0 & 2 \\ 2 & 1 & -3 \end{vmatrix}$

6. (a) $\begin{vmatrix} 0 & 0 & -1 & 3 \\ 0 & 1 & 2 & 1 \\ 2 & -2 & 5 & 2 \\ 3 & 3 & 0 & 0 \end{vmatrix}$

(b) $\begin{vmatrix} 4 & 2 & 0 \\ 1 & 1 & 2 \\ -1 & 3 & 4 \end{vmatrix}$

(c) $\begin{vmatrix} -1 & 2 & -1 \\ 3 & 2 & 1 \\ 1 & 4 & 2 \end{vmatrix}$

7. Verify Theorem 3.10 for the matrix

$$A = \begin{bmatrix} -2 & 3 & 0 \\ 4 & 1 & -3 \\ 2 & 0 & 1 \end{bmatrix}$$

by computing $a_{11}A_{12} + a_{21}A_{22} + a_{31}A_{32}$.

8. Let

$$A = \begin{bmatrix} 2 & 1 & 3 \\ -1 & 2 & 0 \\ 3 & -2 & 1 \end{bmatrix}.$$

(a) Find adj A.

(b) Compute $\det(A)$.

(c) Verify Theorem 3.11; that is, show that

$$A(\text{adj } A) = (\text{adj } A)A = \det(A)I_3.$$

9. Let

$$A = \begin{bmatrix} 6 & 2 & 8 \\ -3 & 4 & 1 \\ 4 & -4 & 5 \end{bmatrix}.$$

(a) Find adj A.

(b) Compute $\det(A)$.

(c) Verify Theorem 3.11; that is, show that

$$A(\text{adj } A) = (\text{adj } A)A = \det(A)I_3.$$

In Exercises 10 through 13, compute the inverses of the matrices, if they exist, using Corollary 3.3.

10. (a) $\begin{bmatrix} 3 & 2 \\ -3 & 4 \end{bmatrix}$ (b) $\begin{bmatrix} 4 & 2 & 2 \\ 0 & 1 & 2 \\ 1 & 0 & 3 \end{bmatrix}$

(c) $\begin{bmatrix} 2 & 0 & -1 \\ 3 & 7 & 2 \\ 1 & 1 & 0 \end{bmatrix}$

11. (a) $\begin{bmatrix} 1 & 2 & -3 \\ -4 & -5 & 2 \\ -1 & 1 & -7 \end{bmatrix}$ (b) $\begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}$

(c) $\begin{bmatrix} 4 & 0 & 2 \\ 0 & 3 & 4 \\ 0 & 1 & -2 \end{bmatrix}$

12. (a) $\begin{bmatrix} 2 & 0 & 1 \\ 3 & 2 & -1 \\ 1 & 0 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 5 & -1 \\ 2 & -1 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 2 & 4 \\ 1 & -5 & 6 \\ 3 & -1 & 2 \end{bmatrix}$

13. (a) $\begin{bmatrix} -3 & 1 \\ 2 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 4 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

(c) $\begin{bmatrix} 0 & 2 & 1 & 3 \\ 2 & -1 & 3 & 4 \\ -2 & 1 & 5 & 2 \\ 0 & 1 & 0 & 2 \end{bmatrix}$

14. Use Theorem 3.12 to determine which of the following matrices are nonsingular.

(a) $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 2 & -3 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 3 & 2 \\ 2 & 1 & 4 \\ 1 & -7 & 2 \end{bmatrix}$

(d) $\begin{bmatrix} 1 & 2 & 0 & 5 \\ 3 & 4 & 1 & 7 \\ -2 & 5 & 2 & 0 \\ 0 & 1 & 2 & -7 \end{bmatrix}$

15. Use Theorem 3.12 to determine which of the following matrices are nonsingular.

(a) $\begin{bmatrix} 4 & 3 & -5 \\ -2 & -1 & 3 \\ 4 & 6 & -2 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 3 & -1 & 2 \\ 2 & -6 & 4 & 1 \\ 3 & 5 & -1 & 3 \\ 4 & -6 & 5 & 2 \end{bmatrix}$

(c) $\begin{bmatrix} 2 & 2 & -4 \\ 1 & 5 & 2 \\ 3 & 7 & -2 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 0 \\ 1 & 3 & 4 \end{bmatrix}$

16. Find all values of λ for which

(a) $\det\left(\begin{bmatrix} \lambda - 2 & 2 \\ 3 & \lambda - 3 \end{bmatrix}\right) = 0$

(b) $\det(\lambda I_3 - A) = 0$, where $A = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & 1 \\ 0 & 0 & -1 \end{bmatrix}$

17. Find all values of λ for which

(a) $\det\left(\begin{bmatrix} \lambda - 1 & -4 \\ 0 & \lambda - 4 \end{bmatrix}\right) = 0$

(b) $\det(\lambda I_3 - A) = 0$, where

$$A = \begin{bmatrix} -3 & -1 & -3 \\ 0 & 3 & 0 \\ -2 & -1 & -2 \end{bmatrix}$$

18. Use Corollary 3.4 to find whether the following homogeneous systems have nontrivial solutions.

(a) $\begin{aligned} x - 2y + z &= 0 \\ 2x + 3y + z &= 0 \\ 3x + y + 2z &= 0 \end{aligned}$

(b) $\begin{aligned} x + 2y + w &= 0 \\ x + 2y + 3z &= 0 \\ z + 2w &= 0 \\ y + 2z - w &= 0 \end{aligned}$

19. Repeat Exercise 18 for the following homogeneous systems.

(a) $\begin{aligned} x + 2y - z &= 0 \\ 2x + y + 2z &= 0 \\ 3x - y + z &= 0 \end{aligned}$

(b) $\begin{aligned} x + y + 2z + w &= 0 \\ 2x - y + z - w &= 0 \\ 3x + y + 2z + 3w &= 0 \\ 2x - y - z + w &= 0 \end{aligned}$

In Exercises 20 through 23, if possible, solve the given linear system by Cramer's rule.

20. $\begin{aligned} 2x + 4y + 6z &= 2 \\ x + 2z &= 0 \\ 2x + 3y - z &= -5 \end{aligned}$

21. $\begin{aligned} x + y + z - 2w &= -4 \\ 2y + z + 3w &= 4 \\ 2x + y - z + 2w &= 5 \\ x - y + w &= 4 \end{aligned}$

22. $\begin{aligned} 2x + y + z &= 6 \\ 3x + 2y - 2z &= -2 \\ x + y + 2z &= 4 \end{aligned}$

23. $\begin{aligned} 2x + 3y + 7z &= 2 \\ -2x - 4z &= 0 \\ x + 2y + 4z &= 0 \end{aligned}$

In Exercises 24 and 25, determine which of the following bit matrices are nonsingular using any of the techniques in the list of nonsingular equivalences.

24. (a) $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$

25. (a) $\begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{bmatrix}$

Theoretical Exercises

T.1. Show by a column (row) expansion that if $A = \left[a_{ij}\right]$ is upper (lower) triangular, then $\det(A) = a_{11}a_{22} \cdots a_{nn}$.

T.2. If $A = \left[a_{ij}\right]$ is a 3×3 matrix, develop the general expression for $\det(A)$ by expanding (a) along the second column, and (b) along the third row. Compare these answers with those obtained for Example 6 in Section 3.1.

T.3. Show that if A is symmetric, then adj A is also symmetric.

T.4. Show that if A is a nonsingular upper triangular matrix, then A^{-1} is also upper triangular.

T.5. Show that

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

is nonsingular if and only if $ad - bc \neq 0$. If this condition is satisfied, use Corollary 3.3 to find A^{-1}.

T.6. Using Corollary 3.3, find the inverse of

$$A = \begin{bmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{bmatrix}.$$

[*Hint*: See Exercise T.12 in Section 3.1, where $\det(A)$ was computed.]

T.7. Show that if A is singular, then adj A is singular. [*Hint*: Show that if A is singular, then $A(\text{adj } A) = O$.]

T.8. Show that if A is an $n \times n$ matrix, then $\det(\text{adj } A) = [\det(A)]^{n-1}$.

T.9. Do Exercise T.10 in Section 1.6 using determinants.

T.10. Let $AB = AC$. Show that if $\det(A) \neq 0$, then $B = C$.

T.11. Let A be an $n \times n$ matrix all of whose entries are integers. Show that if $\det(A) = \pm 1$, then all entries of A^{-1} are integers.

T.12. Show that if A is nonsingular, then adj A is nonsingular and

$$(\text{adj } A)^{-1} = \frac{1}{\det(A)} A = \text{adj } (A^{-1}).$$

T.13. Let A be a 2×2 bit matrix such that A is row equivalent to I_2. Determine all possible such matrices A. (*Hint*: See Exercise T.19 in Section 3.1 or Exercise T.11 in Section 1.6.)

MATLAB Exercises

ML.1. In MATLAB there is a routine **cofactor** that computes the (i, j) cofactor of a matrix. For directions on using this routine, type **help cofactor**. Use **cofactor** to check your hand computations for the matrix A in Exercise 1.

ML.2. Use the **cofactor** routine (see Exercise ML.1) to compute the cofactor of the elements in the second row of

$$A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & -1 & 3 \\ 3 & 2 & 1 \end{bmatrix}.$$

ML.3. Use the **cofactor** routine to evaluate the determinant of A using Theorem 3.9.

$$A = \begin{bmatrix} 4 & 0 & -1 \\ -2 & 2 & -1 \\ 0 & 4 & -3 \end{bmatrix}$$

ML.4. Use the **cofactor** routine to evaluate the determinant of A using Theorem 3.9.

$$A = \begin{bmatrix} -1 & 2 & 0 & 0 \\ 2 & -1 & 2 & 0 \\ 0 & 2 & -1 & 2 \\ 0 & 0 & 2 & -1 \end{bmatrix}$$

ML.5. In MATLAB there is a routine **adjoint**, which computes the adjoint of a matrix. For directions on using this routine, type **help adjoint**. Use **adjoint** to aid in computing the inverses of the matrices in Exercise 11.

3.3 DETERMINANTS FROM A COMPUTATIONAL POINT OF VIEW

In this book we have, by now, developed two methods for solving a linear system of n equations in n unknowns: Gauss–Jordan reduction and Cramer's rule. We also have two methods for finding the inverse of a nonsingular matrix: the method involving determinants and the method discussed in Section 1.7. In this section we discuss criteria to be considered when selecting one or another of these methods.

Most sizable problems in linear algebra are solved on computers so that it is natural to compare two methods by estimating their computing time for the same problem. Since addition is so much faster than multiplication, the number of multiplications is often used as a basis of comparison for two numerical procedures.

Consider the linear system $A\mathbf{x} = \mathbf{b}$, where A is 25×25. If we find $\mathbf{x}$ by Cramer's rule, we must first obtain $\det(A)$. We can find $\det(A)$ by cofactor expansion, say $\det(A) = a_{11}A_{11} + a_{21}A_{21} + \cdots + a_{n1}A_{n1}$, where we have expanded $\det(A)$ along the first column. Note that if each cofactor is available, we require 25 multiplications. Now each cofactor A_{ij} is plus or minus the determinant of a 24×24 matrix, which can be expanded along a given row or column, requiring 24 multiplications. Thus the computation of $\det(A)$ requires more than $25 \times 24 \times \cdots \times 2 \times 1 = 25!$ multiplications. Even if

This linear combination can be written as the matrix product (verify)

$$\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix},$$

which has the form of a linear system. Forming the corresponding augmented matrix and transforming it to reduced row echelon form, we obtain (verify)

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

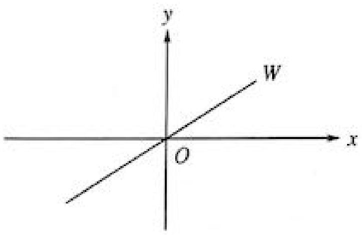
Hence the linear system is consistent with $c_1 = 0$, $c_2 = 1$, and $c_3 = 1$. Thus $\mathbf{u}$ is in span $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$. ■

Key Terms

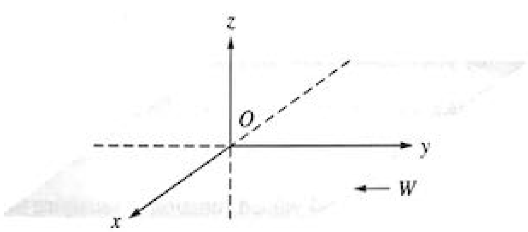
Subspace
Zero subspace
Closure property
Solution space
Linear combination

6.2 Exercises

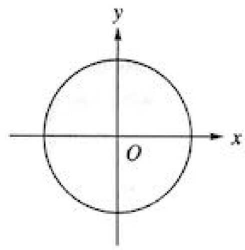
1. The set W consisting of all the points in R^2 of the form (x, x) is a straight line. Is W a subspace of R^2? Explain.



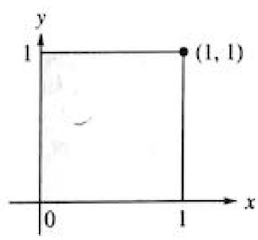
2. Let W be the set of all points in R^3 that lie in the xy-plane. Is W a subspace of R^3? Explain.



3. Consider the circle in the xy-plane centered at the origin whose equation is $x^2 + y^2 = 1$. Let W be the set of all vectors whose tail is at the origin and whose head is a point inside or on the circle. Is W a subspace of R^2? Explain.



4. Consider the unit square shown in the accompanying figure. Let W be the set of all vectors of the form $\begin{bmatrix} x \\ y \end{bmatrix}$, where $0 \leq x \leq 1$, $0 \leq y \leq 1$. That is, W is the set of all vectors whose tail is at the origin and whose head is a point inside or on the square. Is W a subspace of R^2? Explain.



5. Which of the following subsets of R^3 are subspaces of R^3? The set of all vectors of the form
   (a) $(a, b, 2)$
   (b) (a, b, c), where $c = a + b$
   (c) (a, b, c), where $c > 0$

6. Which of the following subsets of R^3 are subspaces of R^3? The set of all vectors of the form
   (a) (a, b, c), where $a = c = 0$
   (b) (a, b, c), where $a = -c$
   (c) (a, b, c), where $b = 2a + 1$

7. Which of the following subsets of R^4 are subspaces of R^4? The set of all vectors of the form
   (a) (a, b, c, d), where $a - b = 2$
   (b) (a, b, c, d), where $c = a + 2b$ and $d = a - 3b$
   (c) (a, b, c, d), where $a = 0$ and $b = -d$

8. Which of the following subsets of R^4 are subspaces of R^4? The set of all vectors of the form
   (a) (a, b, c, d), where $a = b = 0$
   (b) (a, b, c, d), where $a = 1, b = 0$, and $c + d = 1$
   (c) (a, b, c, d), where $a > 0$ and $b < 0$

9. Which of the following subsets of P_2 are subspaces? The set of all polynomials of the form
   (a) $a_2t^2 + a_1t + a_0$, where $a_0 = 0$
   (b) $a_2t^2 + a_1t + a_0$, where $a_0 = 2$
   (c) $a_2t^2 + a_1t + a_0$, where $a_2 + a_1 = a_0$

10. Which of the following subsets of P_2 are subspaces? The set of all polynomials of the form
   (a) $a_2t^2 + a_1t + a_0$, where $a_1 = 0$ and $a_0 = 0$
   (b) $a_2t^2 + a_1t + a_0$, where $a_1 = 2a_0$
   (c) $a_2t^2 + a_1t + a_0$, where $a_2 + a_1 + a_0 = 2$

11. (a) Show that P_2 is a subspace of P_3.
   (b) Show that P_n is a subspace of P_{n+1}.

12. Show that P_n is a subspace of P.

13. Show that P is a subspace of the vector space defined in Example 5 of Section 6.1.

14. Let $\mathbf{u} = (1, 2, -3)$ and $\mathbf{v} = (-2, 3, 0)$ be two vectors in R^3 and let W be the subset of R^3 consisting of all vectors of the form $a\mathbf{u} + b\mathbf{v}$, where a and b are any real numbers. Give an argument to show that W is a subspace of R^3.

15. Let $\mathbf{u} = (2, 0, 3, -4)$ and $\mathbf{v} = (4, 2, -5, 1)$ be two vectors in R^4 and let W be the subset of R^4 consisting of all vectors of the form $a\mathbf{u} + b\mathbf{v}$, where a and b are any real numbers. Give an argument to show that W is a subspace of R^4.

16. Which of the following subsets of the vector space M_{23} defined in Example 4 of Section 6.1 are subspaces? The set of all matrices of the form
   (a) $\begin{bmatrix} a & b & c \\ d & 0 & 0 \end{bmatrix}$, where $b = a + c$
   (b) $\begin{bmatrix} a & b & c \\ d & 0 & 0 \end{bmatrix}$, where $c > 0$
   (c) $\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$, where $a = -2c$ and $f = 2e + d$

17. Which of the following subsets of the vector space M_{23} defined in Example 4 of Section 6.1 are subspaces? The set of all matrices of the form
   (a) $\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$, where $a = 2c + 1$
   (b) $\begin{bmatrix} 0 & 1 & a \\ b & c & 0 \end{bmatrix}$
   (c) $\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$, where $a + c = 0$ and $b + d + f = 0$

18. Which of the following subsets of the vector space M_{nn} are subspaces?
   (a) The set of all $n \times n$ symmetric matrices
   (b) The set of all $n \times n$ nonsingular matrices
   (c) The set of all $n \times n$ diagonal matrices

19. Which of the following subsets of the vector space M_{nn} are subspaces?
   (a) The set of all $n \times n$ singular matrices
   (b) The set of all $n \times n$ upper triangular matrices
   (c) The set of all $n \times n$ matrices whose determinant is 1

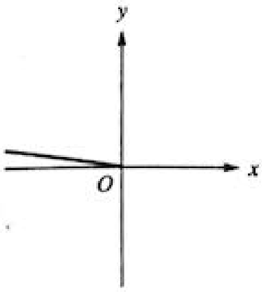
20. (***Calculus Required***) Which of the following subsets are subspaces of the vector space $C(-\infty, \infty)$ defined in Example 8?
   (a) All nonnegative functions
   (b) All constant functions
   (c) All functions f such that $f(0) = 0$
   (d) All functions f such that $f(0) = 5$
   (e) All differentiable functions.

21. (***Calculus Required***) Which of the following subsets of the vector space $C(-\infty, \infty)$ defined in Example 8 are subspaces?
   (a) All integrable functions
   (b) All bounded functions
   (c) All functions that are integrable on $[a, b]$
   (d) All functions that are bounded on $[a, b]$

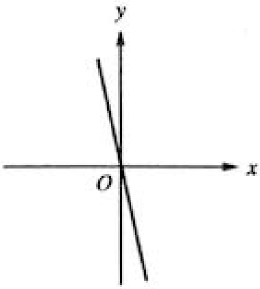
22. (***Calculus Required***) Consider the differential equation
$$y'' - y' + 2y = 0.$$
   A solution is a real-valued function f satisfying the equation. Let V be the set of all solutions to the given differential equation; define $\oplus$ and $\odot$ as in Example 5 in Section 6.1. Show that V is a subspace of the vector space of all real-valued functions defined on $(-\infty, \infty)$. (See also Section 9.2.)

23. Determine which of the following subsets of R^2 are subspaces.

(a)

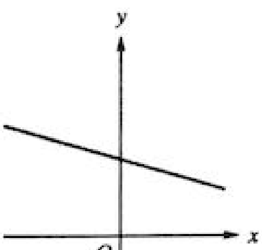


(b)

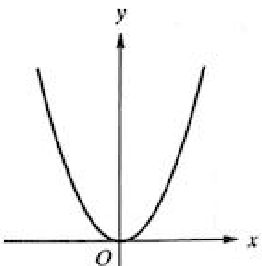


24. Determine which of the following subsets of R^2 are subspaces.

(a)



(b)



25. In each part, determine whether the given vector $\mathbf{v}$ belongs to span $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, where

$$\mathbf{v}_1 = (1, 0, 0, 1), \qquad \mathbf{v}_2 = (1, -1, 0, 0),$$

and

$$\mathbf{v}_3 = (0, 1, 2, 1).$$

(a) $\mathbf{v} = (-1, 4, 2, 2)$ (b) $\mathbf{v} = (1, 2, 0, 1)$

(c) $\mathbf{v} = (-1, 1, 4, 3)$ (d) $\mathbf{v} = (0, 1, 1, 0)$

26. Which of the following vectors are linear combinations of

$$A_1 = \begin{bmatrix} 1 & -1 \\ 0 & 3 \end{bmatrix}, \qquad A_2 = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix},$$

$$A_3 = \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix}?$$

(a) $\begin{bmatrix} 5 & 1 \\ -1 & 9 \end{bmatrix}$ (b) $\begin{bmatrix} -3 & -1 \\ 3 & 2 \end{bmatrix}$

(c) $\begin{bmatrix} 3 & -2 \\ 3 & 2 \end{bmatrix}$ (d) $\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$

27. In each part, determine whether the given vector $p(t)$ belongs to span $\{p_1(t), p_2(t), p_3(t)\}$, where

$$p_1(t) = t^2 - t,$$
$$p_2(t) = t^2 - 2t + 1,$$
$$p_3(t) = -t^2 + 1.$$

(a) $p(t) = 3t^2 - 3t + 1$ (b) $p(t) = t^2 - t + 1$

(c) $p(t) = t + 1$ (d) $p(t) = 2t^2 - t - 1$

Exercises 28 through 33 use bit matrices.

28. Let $V = B^3$. Determine if

$$W = \left\{ \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

is a subspace of V.

29. Let $V = B^3$. Determine if

$$W = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right\}$$

is a subspace of V.

30. Let $V = B^4$. Determine if W, the set of all vectors in V with first entry zero, is a subspace of V.

31. Let $V = B^4$. Determine if W, the set of all vectors in V with second entry one, is a subspace of V.

32. Let

$$S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

Determine if $\mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ belongs to span S.

33. Let

$$S = \left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

Determine if $\mathbf{u} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ belongs to span S.

Theoretical Exercises

T.1. Prove Theorem 6.2.

T.2. Show that a subset W of a vector space V is a subspace of V if and only if the following condition holds: If $\mathbf{u}$ and $\mathbf{v}$ are any vectors in W and a and b are any scalars, then $a\mathbf{u} + b\mathbf{v}$ is in W.

T.3. Show that the set of all solutions to $A\mathbf{x} = \mathbf{b}$, where A is $m \times n$, is not a subspace of R^n if $\mathbf{b} \neq \mathbf{0}$.

T.4. Prove Theorem 6.3.

T.5. Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ be a set of vectors in a vector space V, and let W be a subspace of V containing S. Show that W contains span S.

T.6. If A is a nonsingular matrix, what is the null space of A? Justify your answer.

T.7. Let $\mathbf{x}_0$ be a fixed vector in a vector space V. Show that the set W consisting of all scalar multiples $c\mathbf{x}_0$ of $\mathbf{x}_0$ is a subspace of V.

T.8. Let A be an $m \times n$ matrix. Is the set W of all vectors $\mathbf{x}$ in R^n such that $A\mathbf{x} \neq \mathbf{0}$ a subspace of R^n? Justify your answer.

T.9. Show that the only subspaces of R^1 are $\{\mathbf{0}\}$ and R^1 itself.

T.10. Let W_1 and W_2 be subspaces of a vector space V. Let $W_1 + W_2$ be the set of all vectors $\mathbf{v}$ in V such that $\mathbf{v} = \mathbf{w}_1 + \mathbf{w}_2$, where $\mathbf{w}_1$ is in W_1 and $\mathbf{w}_2$ is in W_2. Show that $W_1 + W_2$ is a subspace of V.

T.11. Let W_1 and W_2 be subspaces of a vector space V with $W_1 \cap W_2 = \{\mathbf{0}\}$. Let $W_1 + W_2$ be as defined in Exercise T.10. Suppose that $V = W_1 + W_2$. Show that every vector in V can be uniquely written as $\mathbf{w}_1 + \mathbf{w}_2$, where $\mathbf{w}_1$ is in W_1 and $\mathbf{w}_2$ is in W_2. In this case we write $V = W_1 \oplus W_2$ and say that V is the **direct sum** of the subspaces W_1 and W_2.

T.12. Show that the set of all points in the plane $ax + by + cz = 0$ is a subspace of R^3.

T.13. Show that if a subset W of a vector space V does not contain the zero vector, then W is not a subspace of V.

T.14. Let $V = B^3$ and $W = \{\mathbf{w}_1\}$, where $\mathbf{w}_1$ is any vector in B^3. Is W a subspace of V?

T.15. Let $V = B^3$. Determine if there is a subspace of V that contains exactly three different vectors.

T.16. In Example 14, W is a subspace of B^3 with exactly four vectors. Determine two other subspaces of B^3 that contain exactly four vectors.

T.17. Determine all the subspaces of B^3 that contain

$$\mathbf{v} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

MATLAB Exercises

ML.1. Let V be R^3 and let W be the subset of V of vectors of the form $(2, a, b)$, where a and b are any real numbers. Is W a subspace of V? Use the following MATLAB commands to help you determine the answer.

```
a1 = fix(10 * randn);
a2 = fix(10 * randn);
b1 = fix(10 * randn);
b2 = fix(10 * randn);
v = [2 a1 b1]
w = [2 a2 b2]
v + w
3 * v
```

ML.2. Let V be P_2 and let W be the subset of V of vectors of the form $ax^2 + bx + 5$, where a and b are arbitrary real numbers. With each such polynomial in W we associate a vector $(a, b, 5)$ in R^3. Construct commands like those in Exercise ML.1 to show that W is not a subspace of V.

Before solving the following MATLAB exercises, you should have read Section 12.7.

ML.3. Use MATLAB to determine if vector $\mathbf{v}$ is a linear combination of the members of set S.

(a) $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$
$= \{(1, 0, 0, 1), (0, 1, 1, 0), (1, 1, 1, 1)\}$
$\mathbf{v} = (0, 1, 1, 1)$

(b) $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$

$$= \left\{ \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 8 \\ -3 \end{bmatrix} \right\}$$

$$\mathbf{v} = \begin{bmatrix} 0 \\ 5 \\ -2 \end{bmatrix}$$

ML.4. Use MATLAB to determine if $\mathbf{v}$ is a linear combination of the members of set S. If it is, express $\mathbf{v}$ in terms of the members of S.

(a) $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$
$= \{(1, 2, 1), (3, 0, 1), (1, 8, 3)\}$
$\mathbf{v} = (-2, 14, 4)$

(b) $S = \{A_1, A_2, A_3\}$

$$= \left\{ \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}, \begin{bmatrix} -3 & 1 \\ 0 & 1 \end{bmatrix} \right\}$$

$\mathbf{v} = I_2$

ML.5. Use MATLAB to determine if $\mathbf{v}$ is a linear combination of the members of set S. If it is, express $\mathbf{v}$ in terms of the members of S.

(a) $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$

$$= \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right\}$$

$$\mathbf{v} = \begin{bmatrix} 0 \\ -1 \\ 1 \\ -2 \\ 1 \end{bmatrix}$$

(b) $S = \{p_1(t), p_2(t), p_3(t)\}$
$= \{2t^2 - t + 1, t^2 - 2, t - 1\}$
$\mathbf{v} = p(t) = 4t^2 + t - 5$

ML.6. In each part, determine whether $\mathbf{v}$ belongs to span S, where

$$S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$$
$$= \{(1, 1, 0, 1), (1, -1, 0, 1), (0, 1, 2, 1)\}.$$

(a) $\mathbf{v} = (2, 3, 2, 3)$

(b) $\mathbf{v} = (2, -3, -2, 3)$

(c) $\mathbf{v} = (0, 1, 2, 3)$

ML.7. In each part, determine whether $p(t)$ belongs to span S, where

$$S = \{p_1(t), p_2(t), p_3(t)\}$$
$$= \{t - 1, t + 1, t^2 + t + 1\}.$$

(a) $p(t) = t^2 + 2t + 4$

(b) $p(t) = 2t^2 + t - 2$

(c) $p(t) = -2t^2 + 1$

6.3 LINEAR INDEPENDENCE

Thus far we have defined a mathematical system called a real vector space and noted some of its properties. We further observe that the only real vector space having a finite number of vectors in it is the vector space whose only vector is $\mathbf{0}$, for if $\mathbf{v} \neq \mathbf{0}$ is in a vector space V, then by Exercise T.4 in Section 6.1, $c\mathbf{v} \neq c'\mathbf{v}$, where c and c' are distinct real numbers, and so V has infinitely many vectors in it. However, in this section and the following one we show that most vector spaces V studied here have a set composed of a finite number of vectors that completely describe V; that is, we can write every vector in V as a linear combination of the vectors in this set. It should be noted that, in general, there is more than one such set describing V. We now turn to a formulation of these ideas.

DEFINITION The vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k$ in a vector space V are said to **span** V if every vector in V is a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k$. Moreover, if $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$, then we also say that the set S **spans** V, or that $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ **spans** V, or that V is **spanned by** S, or in the language of Section 6.2, span $S = V$.

> The procedure to check if the vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k$ span the vector space V is as follows.
>
> ***Step 1.*** Choose an arbitrary vector $\mathbf{v}$ in V.
>
> ***Step 2.*** Determine if $\mathbf{v}$ is a linear combination of the given vectors. If it is, then the given vectors span V. If it is not, they do not span V.

Again, in Step 2, we investigate the consistency of a linear system, but this time for a right side that represents an arbitrary vector in a vector space V.

6.3 Exercises

1. Which of the following vectors span R^2?

(a) $(1, 2), (-1, 1)$

(b) $(0, 0), (1, 1), (-2, -2)$

(c) $(1, 3), (2, -3), (0, 2)$

(d) $(2, 4), (-1, 2)$

2. Which of the following sets of vectors span R^3?

(a) $\{(1, -1, 2), (0, 1, 1)\}$

(b) $\{(1, 2, -1), (6, 3, 0), (4, -1, 2), (2, -5, 4)\}$

(c) $\{(2, 2, 3), (-1, -2, 1), (0, 1, 0)\}$

(d) $\{(1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 1, 1)\}$

3. Which of the following vectors span R^4?

(a) $(1, 0, 0, 1), (0, 1, 0, 0), (1, 1, 1, 1), (1, 1, 1, 0)$

(b) $(1, 2, 1, 0), (1, 1, -1, 0), (0, 0, 0, 1)$

(c) $(6, 4, -2, 4), (2, 0, 0, 1), (3, 2, -1, 2),$
$(5, 6, -3, 2), (0, 4, -2, -1)$

(d) $(1, 1, 0, 0), (1, 2, -1, 1), (0, 0, 1, 1),$
$(2, 1, 2, 1)$

4. Which of the following sets of polynomials span P_2?

(a) $\{t^2 + 1, t^2 + t, t + 1\}$

(b) $\{t^2 + 1, t - 1, t^2 + t\}$

(c) $\{t^2 + 2, 2t^2 - t + 1, t + 2, t^2 + t + 4\}$

(d) $\{t^2 + 2t - 1, t^2 - 1\}$

5. Do the polynomials $t^3 + 2t + 1, t^2 - t + 2, t^3 + 2,$
$-t^3 + t^2 - 5t + 2$ span P_3?

6. Find a set of vectors spanning the solution space of $A\mathbf{x} = \mathbf{0}$, where

$$A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 2 & 3 & 1 \\ 2 & 1 & 3 & 1 \\ 1 & 1 & 2 & 1 \end{bmatrix}.$$

7. Find a set of vectors spanning the null space of

$$A = \begin{bmatrix} 1 & 1 & 2 & -1 \\ 2 & 3 & 6 & -2 \\ -2 & 1 & 2 & 2 \\ 0 & -2 & -4 & 0 \end{bmatrix}.$$

8. Let

$$\mathbf{x}_1 = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 4 \\ -7 \\ -1 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

belong to the solution space of $A\mathbf{x} = \mathbf{0}$. Is $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\}$ linearly independent?

9. Let

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 1 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} 1 \\ 6 \\ 2 \\ 0 \end{bmatrix}$$

belong to the null space of A. Is $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\}$ linearly independent?

10. Which of the following sets of vectors in R^3 are linearly dependent? For those that are, express one vector as a linear combination of the rest.

(a) $\{(1, 2, -1), (3, 2, 5)\}$

(b) $\{(4, 2, 1), (2, 6, -5), (1, -2, 3)\}$

(c) $\{(1, 1, 0), (0, 2, 3), (1, 2, 3), (3, 6, 6)\}$

(d) $\{(1, 2, 3), (1, 1, 1), (1, 0, 1)\}$

11. Consider the vector space R^4. Follow the directions of Exercise 10.

(a) $\{(1, 1, 2, 1), (1, 0, 0, 2), (4, 6, 8, 6), (0, 3, 2, 1)\}$

(b) $\{(1, -2, 3, -1), (-2, 4, -6, 2)\}$

(c) $\{(1, 1, 1, 1), (2, 3, 1, 2), (3, 1, 2, 1), (2, 2, 1, 1)\}$

(d) $\{(4, 2, -1, 3), (6, 5, -5, 1), (2, -1, 3, 5)\}$

12. Consider the vector space P_2. Follow the directions of Exercise 10.

(a) $\{t^2 + 1, t - 2, t + 3\}$

(b) $\{2t^2 + 1, t^2 + 3, t\}$

(c) $\{3t + 1, 3t^2 + 1, 2t^2 + t + 1\}$

(d) $\{t^2 - 4, 5t^2 - 5t - 6, 3t^2 - 5t + 2\}$

13. Consider the vector space M_{22}. Follow the directions of Exercise 10.

(a) $\left\{\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \begin{bmatrix} 0 & 3 \\ 1 & 2 \end{bmatrix}, \begin{bmatrix} 2 & 6 \\ 4 & 6 \end{bmatrix}\right\}$

(b) $\left\{\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix}\right\}$

(c) $\left\{\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}, \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}\right\}$

14. Let V be the vector space of all real-valued continuous functions. Follow the directions of Exercise 10.

(a) $\{\cos t, \sin t, e^t\}$ (b) $\{t, e^t, \sin t\}$

(c) $\{t^2, t, e^t\}$ (d) $\{\cos^2 t, \sin^2 t, \cos 2t\}$

15. For what values of c are the vectors $(-1, 0, -1)$, $(2, 1, 2)$, and $(1, 1, c)$ in R^3 linearly dependent?

16. For what values of λ are the vectors $t + 3$ and $2t + \lambda^2 + 2$ in P_1 linearly dependent?

17. Determine if the vectors $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, and $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ span B^3.

18. Determine if the vectors $\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, and $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ span B^3.

19. Determine if the vectors $\begin{bmatrix}1\\1\\0\\0\end{bmatrix}$, $\begin{bmatrix}0\\1\\1\\0\end{bmatrix}$, $\begin{bmatrix}0\\0\\1\\1\end{bmatrix}$, and $\begin{bmatrix}1\\0\\0\\1\end{bmatrix}$ span B^4.

20. Determine if the vectors in Exercise 17 are linearly independent.

21. Determine if the vectors in Exercise 19 are linearly independent.

22. Show that $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$ in Example 15 are linearly dependent using Theorem 6.4.

Theoretical Exercises

T.1. Show that the vectors $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ in R^n are linearly independent.

T.2. Let S_1 and S_2 be finite subsets of a vector space and let S_1 be a subset of S_2. Show that:

(a) If S_1 is linearly dependent, so is S_2.

(b) If S_2 is linearly independent, so is S_1.

T.3. Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ be a set of vectors in a vector space. Show that S is linearly dependent if and only if one of the vectors in S is a linear combination of all the other vectors in S.

T.4. Suppose that $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is a linearly independent set of vectors in a vector space V. Show that $T = \{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$ is also linearly independent, where $\mathbf{w}_1 = \mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3$, $\mathbf{w}_2 = \mathbf{v}_2 + \mathbf{v}_3$, and $\mathbf{w}_3 = \mathbf{v}_3$.

T.5. Suppose that $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is a linearly independent set of vectors in a vector space V. Is $T = \{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$, where $\mathbf{w}_1 = \mathbf{v}_1 + \mathbf{v}_2$, $\mathbf{w}_2 = \mathbf{v}_1 + \mathbf{v}_3$, $\mathbf{w}_3 = \mathbf{v}_2 + \mathbf{v}_3$, linearly dependent or linearly independent? Justify your answer.

T.6. Suppose that $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is a linearly dependent set of vectors in a vector space V. Is $T = \{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$, where $\mathbf{w}_1 = \mathbf{v}_1$, $\mathbf{w}_2 = \mathbf{v}_1 + \mathbf{v}_2$, $\mathbf{w}_3 = \mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3$, linearly dependent or linearly independent? Justify your answer.

T.7. Let $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$ be vectors in a vector space such that $\{\mathbf{v}_1, \mathbf{v}_2\}$ is linearly independent. Show that if $\mathbf{v}_3$ does not belong to span $\{\mathbf{v}_1, \mathbf{v}_2\}$, then $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly independent.

T.8. Let A be an $m \times n$ matrix in reduced row echelon form. Show that the nonzero rows of A, viewed as vectors in R^n, form a linearly independent set of vectors.

T.9. Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ be a set of vectors in a vector space, and let $T = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$, where each $\mathbf{v}_i$, $i = 1, 2, \dots, m$, is a linear combination of the vectors in S. Show that

$$\mathbf{w} = b_1\mathbf{v}_1 + b_2\mathbf{v}_2 + \cdots + b_m\mathbf{v}_m$$

is a linear combination of the vectors in S.

T.10. Suppose that $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a linearly independent set of vectors in R^n. Show that if A is an $n \times n$ nonsingular matrix, then $\{A\mathbf{v}_1, A\mathbf{v}_2, \dots, A\mathbf{v}_n\}$ is linearly independent.

T.11. Let S_1 and S_2 be finite subsets of a vector space V and let S_1 be a subset of S_2. If S_2 is linearly dependent, show by examples that S_1 may be either linearly dependent or linearly independent.

T.12. Let S_1 and S_2 be finite subsets of a vector space and let S_1 be a subset of S_2. If S_1 is linearly independent, show by examples that S_2 may be either linearly dependent or linearly independent.

T.13. Let $\mathbf{u}$ and $\mathbf{v}$ be nonzero vectors in a vector space V. Show that $\{\mathbf{u}, \mathbf{v}\}$ is linearly dependent if and only if there is a scalar k such that $\mathbf{v} = k\mathbf{u}$. Equivalently, $\{\mathbf{u}, \mathbf{v}\}$ is linearly independent if and only if one of the vectors is not a multiple of the other.

T.14. (*Uses material from Section 5.1*) Let $\mathbf{u}$ and $\mathbf{v}$ be linearly independent vectors in R^3. Show that $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{u} \times \mathbf{v}$ form a basis for R^3. [*Hint*: Form Equation (1) and take the dot product with $\mathbf{u} \times \mathbf{v}$.]

T.15. Let W be a subspace of V spanned by the vectors $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_k$. Is there any vector $\mathbf{v}$ in V such that the span of $\{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_k, \mathbf{v}\}$ will also be W? Describe all such vectors $\mathbf{v}$.

MATLAB Exercises

ML.1. Determine if S is linearly independent or linearly dependent.

(a) $S = \{(1, 0, 0, 1), (0, 1, 1, 0), (1, 1, 1, 1)\}$

(b) $S = \left\{\begin{bmatrix}1 & 2\\1 & 0\end{bmatrix}, \begin{bmatrix}2 & -1\\1 & 2\end{bmatrix}, \begin{bmatrix}-3 & 1\\0 & 1\end{bmatrix}\right\}$

(c) $S = \left\{\begin{bmatrix}1\\2\\1\\0\\1\end{bmatrix}, \begin{bmatrix}0\\1\\2\\-1\\1\end{bmatrix}, \begin{bmatrix}2\\1\\0\\0\\-1\end{bmatrix}, \begin{bmatrix}-2\\1\\1\\1\\1\end{bmatrix}\right\}$

ML.2. Find a spanning set of the solution space of $A\mathbf{x} = \mathbf{0}$, where

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 1 & 1 & 1 & 2 \\ 2 & -1 & 5 & 7 \\ 0 & 2 & -2 & -2 \end{bmatrix}.$$

6.4 BASIS AND DIMENSION

In this section we continue our study of the structure of a vector space V by determining a smallest set of vectors in V that completely describes V.

BASIS

DEFINITION The vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k$ in a vector space V are said to form a **basis** for V if (a) $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k$ span V and (b) $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k$ are linearly independent.

Remark If $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k$ form a basis for a vector space V, then they must be nonzero (see Example 12 in Section 6.3) and distinct and so we write them as a set $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$.

EXAMPLE 1 The vectors $\mathbf{e}_1 = (1, 0)$ and $\mathbf{e}_2 = (0, 1)$ form a basis for R^2, the vectors $\mathbf{e}_1$, $\mathbf{e}_2$, and $\mathbf{e}_3$ form a basis for R^3 and, in general, the vectors $\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n$ form a basis for R^n. Each of these sets of vectors is called the **natural basis** or **standard basis** for R^2, R^3, and R^n, respectively. ■

EXAMPLE 2 Show that the set $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$, where $\mathbf{v}_1 = (1, 0, 1, 0)$, $\mathbf{v}_2 = (0, 1, -1, 2)$, $\mathbf{v}_3 = (0, 2, 2, 1)$, and $\mathbf{v}_4 = (1, 0, 0, 1)$ is a basis for R^4.

Solution To show that S is linearly independent, we form the equation

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 + c_4\mathbf{v}_4 = \mathbf{0}$$

and solve for c_1, c_2, c_3, and c_4. Substituting for $\mathbf{v}_1$, $\mathbf{v}_2$, $\mathbf{v}_3$, and $\mathbf{v}_4$, we obtain the linear system (verify)

$$\begin{aligned} c_1 \qquad\qquad\qquad + c_4 &= 0 \\ c_2 + 2c_3 \qquad &= 0 \\ c_1 - c_2 + 2c_3 \qquad &= 0 \\ 2c_2 + c_3 + c_4 &= 0, \end{aligned}$$

which has as its only solution $c_1 = c_2 = c_3 = c_4 = 0$ (verify), showing that S is linearly independent. Observe that the coefficient matrix of the preceding linear system consists of the vectors $\mathbf{v}_1$, $\mathbf{v}_2$, $\mathbf{v}_3$, and $\mathbf{v}_4$ written in column form.

To show that S spans R^4, we let $\mathbf{v} = (a, b, c, d)$ be any vector in R^4. We now seek constants k_1, k_2, k_3, and k_4 such that

$$k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 + k_4\mathbf{v}_4 = \mathbf{v}.$$

Substituting for $\mathbf{v}_1$, $\mathbf{v}_2$, $\mathbf{v}_3$, $\mathbf{v}_4$, and $\mathbf{v}$, we find a solution (verify) for k_1, k_2, k_3, and k_4 to the resulting linear system for any a, b, c, and d. Hence S spans R^4 and is a basis for R^4. ■

We form the augmented matrix and use row opertions: Add row 1 to row 4, add row 2 to row 3, and add row 2 to row 4, to obtain the equivalent augmented matrix (verify)

$$\left[\begin{array}{cccc:c} 1 & 0 & 1 & 0 & a \\ 0 & 1 & 1 & 0 & b \\ 0 & 0 & 1 & 1 & b+c \\ 0 & 0 & 0 & 0 & a+b+d \end{array}\right].$$

It follows that this system is inconsistent if the choice of bits a, b, and d is such that $a+b+d \neq 0$. For example, if

$$\mathbf{w} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix},$$

then the system is inconsistent; hence S does not span B^4 and is not a basis for B^4. ∎

Key Terms

Basis
Natural (or standard) basis
Finite-dimensional vector space
Infinite-dimensional vector space
Dimension

6.4 Exercises

1. Which of the following sets of vectors are bases for R^2?
   (a) $\{(1, 3), (1, -1)\}$
   (b) $\{(0, 0), (1, 2), (2, 4)\}$
   (c) $\{(1, 2), (2, -3), (3, 2)\}$
   (d) $\{(1, 3), (-2, 6)\}$

2. Which of the following sets of vectors are bases for R^3?
   (a) $\{(1, 2, 0), (0, 1, -1)\}$
   (b) $\{(1, 1, -1), (2, 3, 4), (4, 1, -1), (0, 1, -1)\}$
   (c) $\{(3, 2, 2), (-1, 2, 1), (0, 1, 0)\}$
   (d) $\{(1, 0, 0), (0, 2, -1), (3, 4, 1), (0, 1, 0)\}$

3. Which of the following sets of vectors are bases for R^4?
   (a) $\{(1, 0, 0, 1), (0, 1, 0, 0), (1, 1, 1, 1), (0, 1, 1, 1)\}$
   (b) $\{(1, -1, 0, 2), (3, -1, 2, 1), (1, 0, 0, 1)\}$
   (c) $\{(-2, 4, 6, 4), (0, 1, 2, 0), (-1, 2, 3, 2), (-3, 2, 5, 6), (-2, -1, 0, 4)\}$
   (d) $\{(0, 0, 1, 1), (-1, 1, 1, 2), (1, 1, 0, 0), (2, 1, 2, 1)\}$

4. Which of the following sets of vectors are bases for P_2?
   (a) $\{-t^2 + t + 2, 2t^2 + 2t + 3, 4t^2 - 1\}$
   (b) $\{t^2 + 2t - 1, 2t^2 + 3t - 2\}$
   (c) $\{t^2 + 1, 3t^2 + 2t, 3t^2 + 2t + 1, 6t^2 + 6t + 3\}$
   (d) $\{3t^2 + 2t + 1, t^2 + t + 1, t^2 + 1\}$

5. Which of the following sets of vectors are bases for P_3?
   (a) $\{t^3 + 2t^2 + 3t, 2t^3 + 1, 6t^3 + 8t^2 + 6t + 4, t^3 + 2t^2 + t + 1\}$
   (b) $\{t^3 + t^2 + 1, t^3 - 1, t^3 + t^2 + t\}$
   (c) $\{t^3 + t^2 + t + 1, t^3 + 2t^2 + t + 3, 2t^3 + t^2 + 3t + 2, t^3 + t^2 + 2t + 2\}$
   (d) $\{t^3 - t, t^3 + t^2 + 1, t - 1\}$

6. Show that the matrices

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$$

form a basis for the vector space M_{22}.

In Exercises 7 and 8, determine which of the given subsets forms a basis for R^3. Express the vector $(2, 1, 3)$ as a linear combination of the vectors in each subset that is a basis.

7. (a) $\{(1, 1, 1), (1, 2, 3), (0, 1, 0)\}$
   (b) $\{(1, 2, 3), (2, 1, 3), (0, 0, 0)\}$

8. (a) $\{(2, 1, 3), (1, 2, 1), (1, 1, 4), (1, 5, 1)\}$
   (b) $\{(1, 1, 2), (2, 2, 0), (3, 4, -1)\}$

In Exercises 9 and 10, determine which of the given subsets forms a basis for P_2. Express $5t^2 - 3t + 8$ as a linear combination of the vectors in each subset that is a basis.

9. (a) $\{t^2 + t, t - 1, t + 1\}$

(b) $\{t^2 + 1, t - 1\}$

10. (a) $\{t^2 + t, t^2, t^2 + 1\}$

(b) $\{t^2 + 1, t^2 - t + 1\}$

11. Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$, where

$$\mathbf{v}_1 = (1, 2, 2), \quad \mathbf{v}_2 = (3, 2, 1),$$
$$\mathbf{v}_3 = (11, 10, 7), \quad \text{and} \quad \mathbf{v}_4 = (7, 6, 4).$$

Find a basis for the subspace $W = \operatorname{span} S$ of R^3. What is $\dim W$?

12. Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4, \mathbf{v}_5\}$, where

$$\mathbf{v}_1 = (1, 1, 0, -1), \quad \mathbf{v}_2 = (0, 1, 2, 1),$$
$$\mathbf{v}_3 = (1, 0, 1, -1), \quad \mathbf{v}_4 = (1, 1, -6, -3),$$

and $\mathbf{v}_5 = (-1, -5, 1, 0)$. Find a basis for the subspace $W = \operatorname{span} S$ of R^4. What is $\dim W$?

13. Consider the following subset of P_3:

$$S = \{t^3 + t^2 - 2t + 1, t^2 + 1, t^3 - 2t,$$
$$2t^3 + 3t^2 - 4t + 3\}.$$

Find a basis for the subspace $W = \operatorname{span} S$. What is $\dim W$?

14. Let

$$S = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \right\}.$$

Find a basis for the subspace $W = \operatorname{span} S$ of M_{22}.

15. Find a basis for M_{23}. What is the dimension of M_{23}? Generalize to M_{mn}.

16. Consider the following subset of the vector space of all real-valued functions

$$S = \{\cos^2 t, \sin^2 t, \cos 2t\}.$$

Find a basis for the subspace $W = \operatorname{span} S$. What is $\dim W$?

In Exercises 17 and 18, find a basis for the given subspaces of R^3 and R^4.

17. (a) All vectors of the form (a, b, c), where $b = a + c$

(b) All vectors of the form (a, b, c), where $b = a$

(c) All vectors of the form (a, b, c), where $2a + b - c = 0$

18. (a) All vectors of the form (a, b, c), where $a = 0$

(b) All vectors of the form $(a + c, a - b, b + c, -a + b)$

(c) All vectors of the form (a, b, c), where $a - b + 5c = 0$

In Exercises 19 and 20, find the dimensions of the given subspaces of R^4.

19. (a) All vectors of the form (a, b, c, d), where $d = a + b$

(b) All vectors of the form (a, b, c, d), where $c = a - b$ and $d = a + b$

20. (a) All vectors of the form (a, b, c, d), where $a = b$

(b) All vectors of the form $(a + c, -a + b, -b - c, a + b + 2c)$

21. Find a basis for the subspace of P_2 consisting of all vectors of the form $at^2 + bt + c$, where $c = 2a - 3b$.

22. Find a basis for the subspace of P_3 consisting of all vectors of the form $at^3 + bt^2 + ct + d$, where $a = b$ and $c = d$.

23. Find the dimensions of the subspaces of R^2 spanned by the vectors in Exercise 1.

24. Find the dimensions of the subspaces of R^3 spanned by the vectors in Exercise 2.

25. Find the dimensions of the subspaces of R^4 spanned by the vectors in Exercise 3.

26. Find the dimension of the subspace of P_2 consisting of all vectors of the form $at^2 + bt + c$, where $c = b - 2a$.

27. Find the dimension of the subspace of P_3 consisting of all vectors of the form $at^3 + bt^2 + ct + d$, where $b = 3a - 5d$ and $c = d + 4a$.

28. Find a basis for R^3 that includes the vectors

(a) $(1, 0, 2)$

(b) $(1, 0, 2)$ and $(0, 1, 3)$

29. Find a basis for R^4 that includes the vectors $(1, 0, 1, 0)$ and $(0, 1, -1, 0)$.

30. Find all values of a for which $\{(a^2, 0, 1), (0, a, 2), (1, 0, 1)\}$ is a basis for R^3.

31. Find a basis for the subspace W of M_{33} consisting of all symmetric matrices.

32. Find a basis for the subspace of M_{33} consisting of all diagonal matrices.

33. Give an example of a two-dimensional subspace of R^4.

34. Give an example of a two-dimensional subspace of P_3.

In Exercises 35 and 36, find a basis for the given plane.

35. $2x - 3y + 4z = 0$. **36.** $x + y - 3z = 0$.

37. Determine if the vectors

$$\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

are a basis for B^3.

38. Determine if the vectors

$$\begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\1\end{bmatrix}$$

are a basis for B^3.

39. Determine if the vectors

$$\begin{bmatrix}1\\0\\0\\1\end{bmatrix}, \begin{bmatrix}0\\0\\1\\1\end{bmatrix}, \begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}1\\1\\0\\0\end{bmatrix}$$

are a basis for B^4.

40. Determine if the vectors

$$\begin{bmatrix}1\\1\\0\\1\end{bmatrix}, \begin{bmatrix}1\\0\\1\\1\end{bmatrix}, \begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}1\\1\\1\\1\end{bmatrix}$$

are a basis for B^4.

Theoretical Exercises

T.1. Suppose that in the nonzero vector space V, the largest number of vectors in a linearly independent set is m. Show that any set of m linearly independent vectors in V is a basis for V.

T.2. Show that if V is a finite-dimensional vector space, then every nonzero subspace W of V has a finite basis and $\dim W \leq \dim V$.

T.3. Show that if $\dim V = n$, then any $n + 1$ vectors in V are linearly dependent.

T.4. Show that if $\dim V = n$, then no set of $n - 1$ vectors in V can span V.

T.5. Prove Theorem 6.8.

T.6. Prove Theorem 6.9.

T.7. Show that if W is a subspace of a finite-dimensional vector space V and $\dim W = \dim V$, then $W = V$.

T.8. Show that the subspaces of R^3 are $\{\mathbf{0}\}$, R^3, all lines through the origin, and all planes through the origin.

T.9. Show that if $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a basis for a vector space V and $c \neq 0$, then $\{c\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is also a basis for V.

T.10. Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ be a basis for vector space V. Then show that $T = \{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$, where

$$\mathbf{w}_1 = \mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3,$$

$$\mathbf{w}_2 = \mathbf{v}_2 + \mathbf{v}_3,$$

and

$$\mathbf{w}_3 = \mathbf{v}_3,$$

is also a basis for V.

T.11. Let

$$S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$$

be a set of nonzero vectors in a vector space V such that every vector in V can be written in one and only one way as a linear combination of the vectors in S. Show that S is a basis for V.

T.12. Suppose that

$$\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$$

is a basis for R^n. Show that if A is an $n \times n$ nonsingular matrix, then

$$\{A\mathbf{v}_1, A\mathbf{v}_2, \dots, A\mathbf{v}_n\}$$

is also a basis for R^n. (*Hint*: See Exercise T.10 in Section 6.3.)

T.13. Suppose that

$$\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$$

is a linearly independent set of vectors in R^n and let A be a singular matrix. Prove or disprove that

$$\{A\mathbf{v}_1, A\mathbf{v}_2, \dots, A\mathbf{v}_n\}$$

is linearly independent.

T.14. Show that the vector space P of all polynomials is not finite-dimensional. [*Hint*: Suppose that

$$\{p_1(t), p_2(t), \dots, p_k(t)\}$$

is a finite basis for P. Let $d_j = \text{degree } p_j(t)$. Establish a contradiction.]

T.15. Show that the set of vectors

$$\{t^n, t^{n-1}, \dots, t, 1\}$$

in P_n is linearly independent.

T.16. Show that if the sum of the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ from B^n is $\mathbf{0}$, then these vectors cannot form a basis for B^n.

T.17. Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ be a set of vectors in B^3.

(a) Find linearly independent vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ such that $\mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3 \neq \mathbf{0}$.

(b) Find linearly dependent vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ such that $\mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3 \neq \mathbf{0}$.

MATLAB Exercises

In order to use MATLAB *in this section, you should have read Section 12.7. In the following exercises we relate the theory developed in the section to computational procedures in* MATLAB *that aid in analyzing the situation.*

To determine if a set $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ is a basis for a vector space V, the definition requires that we show span $S = V$ and S is linearly independent. However, if we know that dim $V = k$*, then Theorem 6.9 implies that we need only show that either span $S = V$ or S is linearly independent. The linear independence, in this special case, is easily analyzed using* MATLAB*'s* **rref** *command. Construct the homogeneous system $\mathbf{Ax} = \mathbf{0}$ associated with the linear independence/dependence question. Then S is linearly independent if and only if*

$$\textbf{rref(A)} = \begin{bmatrix} \mathbf{I}_k \\ \mathbf{0} \end{bmatrix}.$$

In Exercises ML.1 through ML.6, if this special case can be applied, do so; otherwise, determine if S is a basis for V in the conventional manner.

ML.1. $S = \{(1, 2, 1), (2, 1, 1), (2, 2, 1)\}$ in $V = R^3$

ML.2. $S = \{2t - 2, t^2 - 3t + 1, 2t^2 - 8t + 4\}$ in $V = P_2$

ML.3. $S = \{(1, 1, 0, 0), (2, 1, 1, -1), (0, 0, 1, 1), (1, 2, 1, 2)\}$ in $V = R^4$

ML.4. $S = \{(1, 2, 1, 0), (2, 1, 3, 1), (2, -2, 4, 2)\}$ in $V =$ span S

ML.5. $S = \{(1, 2, 1, 0), (2, 1, 3, 1), (2, 2, 1, 2)\}$ in $V =$ span S

ML.6. $V =$ the subspace of R^3 of all vectors of the form (a, b, c), where $b = 2a - c$ and $S = \{(0, 1, -1), (1, 1, 1)\}$.

In Exercises ML.7 through ML.9, use MATLAB*'s* **rref** *command to determine a subset of S that is a basis for span S. See Example 5.*

ML.7. $S = \{(1, 1, 0, 0), (-2, -2, 0, 0), (1, 0, 2, 1), (2, 1, 2, 1), (0, 1, 1, 1)\}$.
What is dim span S? Does span $S = R^4$?

ML.8. $S = \left\{ \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 2 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 4 \\ 2 & 4 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right\}$.
What is dim span S? Does span $S = M_{22}$?

ML.9. $S = \{t - 2, 2t - 1, 4t - 2, t^2 - t + 1, t^2 + 2t + 1\}$.
What is dim span S? Does span $S = P_2$?

An interpretation of Theorem 6.8 is that any linearly independent subset S of vector space V can be extended to a basis for V. Following the ideas in Example 9, use MATLAB*'s* **rref** *command to extend S to a basis for V in Exercises ML.10 through ML.12.*

ML.10. $S = \{(1, 1, 0, 0), (1, 0, 1, 0)\}$, $V = R^4$

ML.11. $S = \{t^3 - t + 1, t^3 + 2\}$, $V = P_3$

ML.12. $S = \{(0, 3, 0, 2, -1)\}$,
$V =$ the subspace of R^5 consisting of all vectors of the form (a, b, c, d, e), where $c = a$, $b = 2d + e$

6.5 HOMOGENEOUS SYSTEMS

Homogeneous systems play a central role in linear algebra. This will be seen in Chapter 8, where the foundations of the subject are all integrated to solve one of the major problems occurring in a wide variety of applications. In this section we deal with several problems involving homogeneous systems that will be fundamental in Chapter 8. Here we are able to focus our attention on these problems without being distracted by the additional material in Chapter 8.

Consider the homogeneous system

$$A\mathbf{x} = \mathbf{0},$$

where A is an $m \times n$ matrix. As we have already observed in Example 9 of Section 6.2, the set of all solutions to this homogeneous system is a subspace of R^n. An extremely important problem, which will occur repeatedly in Chapter 8, is that of finding a basis for this solution space. To find such a basis, we use the method of Gauss–Jordan reduction presented in Section 1.6. Thus we transform the augmented matrix $\left[A \;\vdots\; \mathbf{0}\right]$ of the system to a matrix $\left[B \;\vdots\; \mathbf{0}\right]$ in reduced row echelon form, where B has r nonzero rows, $1 \leq r \leq m$. Without

6.5 Exercises

1. Let
$$A = \begin{bmatrix} 2 & -1 & -2 \\ -4 & 2 & 4 \\ -8 & 4 & 8 \end{bmatrix}.$$

(a) Find the set of all solutions to $A\mathbf{x} = \mathbf{0}$.

(b) Express each solution as a linear combination of two vectors in R^3.

(c) Sketch these vectors in a three-dimensional coordinate system to show that the solution space is a plane through the origin.

2. Let
$$A = \begin{bmatrix} 1 & 1 & -2 \\ -2 & -2 & 4 \\ -1 & -1 & 2 \end{bmatrix}.$$

(a) Find the set of all solutions to $A\mathbf{x} = \mathbf{0}$.

(b) Express each solution as a linear combination of two vectors in R^3.

(c) Sketch these vectors in a three-dimensional coordinate system to show that the solution space is a plane through the origin.

In Exercises 3 through 10, find a basis for and the dimension of the solution space of the given homogeneous system.

3. $$\begin{aligned} x_1 + x_2 + x_3 + x_4 &= 0 \\ 2x_1 + x_2 - x_3 + x_4 &= 0 \end{aligned}$$

4. $$\begin{bmatrix} 1 & -1 & 1 & -2 & 1 \\ 3 & -3 & 2 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

5. $$\begin{aligned} x_1 + 2x_2 - x_3 + 3x_4 &= 0 \\ 2x_1 + 2x_2 - x_3 + 2x_4 &= 0 \\ x_1 \qquad + 3x_3 + 3x_4 &= 0 \end{aligned}$$

6. $$\begin{aligned} x_1 - x_2 + 2x_3 + 3x_4 + 4x_5 &= 0 \\ -x_1 + 2x_2 + 3x_3 + 4x_4 + 5x_5 &= 0 \\ x_1 - x_2 + 3x_3 + 5x_4 + 6x_5 &= 0 \\ 3x_1 - 4x_2 + x_3 + 2x_4 + 3x_5 &= 0 \end{aligned}$$

7. $$\begin{bmatrix} 1 & 2 & 1 & 2 & 1 \\ 1 & 2 & 2 & 1 & 2 \\ 2 & 4 & 3 & 3 & 3 \\ 0 & 0 & 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

8. $$\begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

9. $$\begin{bmatrix} 1 & 2 & 2 & -1 & 1 \\ 0 & 2 & 2 & -2 & -1 \\ 2 & 6 & 2 & -4 & 1 \\ 1 & 4 & 0 & -3 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

10. $$\begin{bmatrix} 1 & 2 & -3 & -2 & 1 & 3 \\ 1 & 2 & -4 & 3 & 3 & 4 \\ -2 & -4 & 6 & 4 & -3 & 2 \\ 0 & 0 & -1 & 5 & 1 & 9 \\ 1 & 2 & -3 & -2 & 0 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

In Exercises 11 and 12, find a basis for the null space of the given matrix A.

11. $$A = \begin{bmatrix} 1 & 2 & 3 & -1 \\ 2 & 3 & 2 & 0 \\ 3 & 4 & 1 & 1 \\ 1 & 1 & -1 & 1 \end{bmatrix}$$

12. $$A = \begin{bmatrix} 1 & -1 & 2 & 1 & 0 \\ 2 & 0 & 1 & -1 & 3 \\ 5 & -1 & 3 & 0 & 3 \\ 4 & -2 & 5 & 1 & 3 \\ 1 & 3 & -4 & -5 & 6 \end{bmatrix}$$

In Exercises 13 through 16, find a basis for the solution space of the homogeneous system $(\lambda I_n - A)\mathbf{x} = \mathbf{0}$ for the given scalar λ and given matrix A.

13. $\lambda = 1$, $A = \begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix}$

14. $\lambda = -3$, $A = \begin{bmatrix} -4 & -3 \\ 2 & 3 \end{bmatrix}$

15. $\lambda = 1$, $A = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & -3 \\ 0 & 1 & 3 \end{bmatrix}$

16. $\lambda = 3$, $A = \begin{bmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix}$

In Exercises 17 through 20, find all real numbers λ such that the homogeneous system $(\lambda I_n - A)\mathbf{x} = \mathbf{0}$ has a nontrivial solution.

17. $A = \begin{bmatrix} 2 & 3 \\ 2 & -3 \end{bmatrix}$

18. $A = \begin{bmatrix} 3 & 0 \\ 2 & -2 \end{bmatrix}$

19. $A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & 0 \end{bmatrix}$

20. $A = \begin{bmatrix} -2 & 0 & 0 \\ 0 & -2 & -3 \\ 0 & 4 & 5 \end{bmatrix}$

In Exercises 21 and 22, solve the given linear system and write the solution $\mathbf{x}$ as $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h$, where $\mathbf{x}_p$ is a particular solution to the given system and $\mathbf{x}_h$ is a solution to the associated homogeneous system.

21. $$\begin{aligned} x + 2y - z - w &= 3 \\ x + y + 3z + 2w &= -2 \\ 2x - y + 4z + 3w &= 1 \\ 2x - 2y + 8z + 6w &= -4 \end{aligned}$$

22. $$\begin{aligned} x - y + 2z + 2w &= 1 \\ -x + 2y + 3z + 2w &= 0 \\ 2x + 2y + z \qquad &= 4 \end{aligned}$$

In Exercises 23 through 26, find a basis for and the dimension of the solution space of the given bit homogeneous system.

23. $$\begin{aligned} x_1 + x_2 + x_4 &= 0 \\ x_1 + x_3 + x_4 &= 0 \\ x_1 + x_2 + x_3 &= 0 \end{aligned}$$

24. $$\begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

25. $$\begin{bmatrix} 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

26. $$\begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

In Exercises 27 and 28, solve the given bit system and write the solution $\mathbf{x}$ *as* $\mathbf{x}_p + \mathbf{x}_h$.

27. $$\begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

28. $$\begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

Theoretical Exercises

T.1. Let $S = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k\}$ be a set of solutions to a homogeneous system $A\mathbf{x} = \mathbf{0}$. Show that every vector in span S is a solution to $A\mathbf{x} = \mathbf{0}$.

T.2. Show that if the $n \times n$ coefficient matrix A of the homogeneous system $A\mathbf{x} = \mathbf{0}$ has a row or column of zeros, then $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution.

T.3. (a) Show that the zero matrix is the only 3×3 matrix whose null space has dimension 3.

(b) Let A be a nonzero 3×3 matrix and suppose that $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution. Show that the dimension of the null space of A is either 1 or 2.

T.4. Matrices A and B are $m \times n$ and their reduced row echelon forms are the same. What is the relationship between the null space of A and the null space of B?

MATLAB Exercises

In Exercises ML.1 through ML.3, use MATLAB*'s* **rref** *command to aid in finding a basis for the null space of A. You may also use routine* **homsoln**. *For directions, use* **help**.

ML.1. $$A = \begin{bmatrix} 1 & 1 & 2 & 2 & 1 \\ 2 & 0 & 4 & 2 & 4 \\ 1 & 1 & 2 & 2 & 1 \end{bmatrix}$$

ML.2. $$A = \begin{bmatrix} 2 & 2 & 2 \\ 1 & 2 & 1 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

ML.3. $$A = \begin{bmatrix} 1 & 4 & 7 & 0 \\ 2 & 5 & 8 & -1 \\ 3 & 6 & 9 & -2 \end{bmatrix}$$

ML.4. For the matrix
$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$
and $\lambda = 3$, the homogeneous system $(\lambda I_2 - A)\mathbf{x} = \mathbf{0}$ has a nontrivial solution. Find such a solution using MATLAB commands.

ML.5. For the matrix
$$A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$
and $\lambda = 6$, the homogeneous linear system $(\lambda I_3 - A)\mathbf{x} = \mathbf{0}$ has a nontrivial solution. Find such a solution using MATLAB commands.

6.6 THE RANK OF A MATRIX AND APPLICATIONS

In this section we obtain another effective method for finding a basis for a vector space V spanned by a given set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$. In the proof of Theorem 6.6 we developed a technique for choosing a basis for V that is a *subset* of S. The method to be developed in this section produces a basis for V that is *not* guaranteed to be a subset of S. We shall also attach a unique number to a matrix A that we later show gives us information about

EXAMPLE 9 Consider the linear system

$$\begin{bmatrix} 2 & 1 & 3 \\ 1 & -2 & 2 \\ 0 & 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

Since rank $A = \text{rank}\left[A \mid \mathbf{b}\right] = 3$, the linear system has a solution. ■

EXAMPLE 10 The linear system

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & -3 & 4 \\ 2 & -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

has no solution because rank $A = 2$ and rank $\left[A \mid \mathbf{b}\right] = 3$ (verify). ■

We now extend our list of nonsingular equivalences.

List of Nonsingular Equivalences

The following statements are equivalent for an $n \times n$ matrix A.

1. A is nonsingular.
2. $\mathbf{x} = \mathbf{0}$ is the only solution to $A\mathbf{x} = \mathbf{0}$.
3. A is row equivalent to I_n.
4. The linear system $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $n \times 1$ matrix $\mathbf{b}$.
5. $\det(A) \neq 0$.
6. A has rank n.
7. A has nullity 0.
8. The rows of A form a linearly independent set of n vectors in R^n.
9. The columns of A form a linearly independent set of n vectors in R^n.

Key Terms

Row space
Column space
Row rank
Column rank
Rank
Nonhomogeneous system

6.6 Exercises

1. Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4, \mathbf{v}_5\}$, where

$$\mathbf{v}_1 = (1, 2, 3), \qquad \mathbf{v}_2 = (2, 1, 4),$$
$$\mathbf{v}_3 = (-1, -1, 2), \qquad \mathbf{v}_4 = (0, 1, 2),$$

and $\mathbf{v}_5 = (1, 1, 1)$. Find a basis for the subspace $V = \text{span } S$ of R^3.

2. Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4, \mathbf{v}_5\}$, where

$$\mathbf{v}_1 = (1, 1, 2, 1), \qquad \mathbf{v}_2 = (1, 0, -3, 1),$$
$$\mathbf{v}_3 = (0, 1, 1, 2), \qquad \mathbf{v}_4 = (0, 0, 1, 1),$$

and $\mathbf{v}_5 = (1, 0, 0, 1)$. Find a basis for the subspace $V = \text{span } S$ of R^4.

3. Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4, \mathbf{v}_5\}$, where

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 2 \end{bmatrix}, \qquad \mathbf{v}_2 = \begin{bmatrix} 2 \\ 1 \\ 2 \\ 1 \end{bmatrix},$$

$$\mathbf{v}_3 = \begin{bmatrix} 3 \\ 2 \\ 3 \\ 2 \end{bmatrix}, \quad \mathbf{v}_4 = \begin{bmatrix} 3 \\ 3 \\ 3 \\ 3 \end{bmatrix}, \quad \text{and} \quad \mathbf{v}_5 = \begin{bmatrix} 5 \\ 3 \\ 5 \\ 3 \end{bmatrix}.$$

Find a basis for the subspace $V = \text{span } S$ of R^4.

4. Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4, \mathbf{v}_5\}$, where

$$\mathbf{v}_1 = \begin{bmatrix}1\\2\\1\\1\end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix}2\\1\\3\\1\end{bmatrix},$$

$$\mathbf{v}_3 = \begin{bmatrix}0\\2\\1\\2\end{bmatrix}, \quad \mathbf{v}_4 = \begin{bmatrix}3\\2\\1\\4\end{bmatrix}, \quad \text{and} \quad \mathbf{v}_5 = \begin{bmatrix}5\\0\\0\\-1\end{bmatrix}.$$

Find a basis for the subspace $V = \operatorname{span} S$ of R^4.

In Exercises 5 and 6, find a basis for the row space of A
(a) *consisting of vectors that are* not *row vectors of A;*
(b) *consisting of vectors that are row vectors of A.*

5. $A = \begin{bmatrix}1 & 2 & -1\\1 & 9 & -1\\-3 & 8 & 3\\-2 & 3 & 2\end{bmatrix}$

6. $A = \begin{bmatrix}1 & 2 & -1 & 3\\3 & 5 & 2 & 0\\0 & 1 & 2 & 1\\-1 & 0 & -2 & 7\end{bmatrix}$

In Exercises 7 and 8, find a basis for the column space of A
(a) *consisting of vectors that are* not *column vectors of A;*
(b) *consisting of vectors that are column vectors of A.*

7. $A = \begin{bmatrix}1 & -2 & 7 & 0\\1 & -1 & 4 & 0\\3 & 2 & -3 & 5\\2 & 1 & -1 & 3\end{bmatrix}$

8. $A = \begin{bmatrix}-2 & 2 & 3 & 7 & 1\\-2 & 2 & 4 & 8 & 0\\-3 & 3 & 2 & 8 & 4\\4 & -2 & 1 & -5 & -7\end{bmatrix}$

In Exercises 9 and 10, compute a basis for the row space of A, the column space of A, the row space of A^T, and the column space of A^T. Write a short description giving the relationships among these bases.

9. $A = \begin{bmatrix}1 & -2 & 5\\2 & 3 & 2\\0 & -7 & 8\end{bmatrix}$

10. $A = \begin{bmatrix}2 & -3 & -7 & 11\\3 & -1 & -7 & 13\\1 & 2 & 0 & 2\end{bmatrix}$

In Exercises 11 and 12, compute the row and column ranks of A, verifying Theorem 6.11.

11. $A = \begin{bmatrix}1 & 2 & 3 & 2 & 1\\3 & 1 & -5 & -2 & 1\\7 & 8 & -1 & 2 & 5\end{bmatrix}$

12. $A = \begin{bmatrix}1 & 3 & 2 & 0 & 0 & 1\\2 & 1 & -5 & 1 & 2 & 0\\3 & 2 & 5 & 1 & -2 & 1\\5 & 8 & 9 & 1 & -2 & 2\\9 & 9 & 4 & 2 & 0 & 2\end{bmatrix}$

In Exercises 13 through 17, compute the rank and nullity of A and verify Theorem 6.12.

13. $A = \begin{bmatrix}1 & 2 & 1 & 3\\2 & 1 & -4 & -5\\7 & 8 & -5 & -1\\10 & 14 & -2 & 8\end{bmatrix}$

14. $A = \begin{bmatrix}1 & 2 & 1 & 3\\2 & 1 & -4 & -5\\1 & 1 & 0 & 0\\0 & 0 & 1 & 1\end{bmatrix}$

15. $A = \begin{bmatrix}1 & 2 & 3\\-1 & 2 & 1\\3 & 1 & 2\end{bmatrix}$ **16.** $A = \begin{bmatrix}1 & -2 & -1\\2 & -1 & 3\\7 & -8 & 3\end{bmatrix}$

17. $A = \begin{bmatrix}1 & -2 & -1\\2 & -1 & 3\\7 & -8 & 3\\5 & -7 & 0\end{bmatrix}$

18. If A is a 3×4 matrix, what is the largest possible value for rank A?

19. If A is a 4×6 matrix, show that the columns of A are linearly dependent.

20. If A is a 5×3 matrix, show that the rows of A are linearly dependent.

In Exercises 21 and 22, let A be a 7×3 matrix whose rank is 3.

21. Are the rows of A linearly dependent or linearly independent? Justify your answer.

22. Are the columns of A linearly dependent or linearly independent? Justify your answer.

In Exercises 23 through 25, use Theorem 6.13 to determine whether each matrix is singular or nonsingular.

23. $\begin{bmatrix}1 & 2 & -3\\-1 & 2 & 3\\0 & 8 & 0\end{bmatrix}$ **24.** $\begin{bmatrix}1 & 2 & -3\\-1 & 2 & 3\\0 & 1 & 1\end{bmatrix}$

25. $\begin{bmatrix}1 & 1 & 4 & -1\\1 & 2 & 3 & 2\\-1 & 3 & 2 & 1\\-2 & 6 & 12 & -4\end{bmatrix}$

In Exercises 26 and 27, use Corollary 6.3 to determine whether the linear system $A\mathbf{x} = \mathbf{b}$ has a unique solution for every 3×1 matrix $\mathbf{b}$.

26. $A = \begin{bmatrix} 1 & 2 & -2 \\ 0 & 8 & -7 \\ 3 & -2 & 1 \end{bmatrix}$

27. $A = \begin{bmatrix} 1 & -1 & 2 \\ 3 & 2 & 3 \\ 1 & -2 & 1 \end{bmatrix}$

Use Corollary 6.4 to do Exercises 28 and 29.

28. Is
$$S = \left\{ \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix} \right\}$$
a linearly independent set of vectors in R^3?

29. Is
$$S = \left\{ \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \\ -5 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} \right\}$$
a linearly independent set of vectors in R^3?

In Exercises 30 through 32, find which homogeneous systems have a nontrivial solution for the given matrix A by using Corollary 6.5.

30. $A = \begin{bmatrix} 1 & 1 & 2 & -1 \\ 1 & 3 & -1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 2 & 1 & 1 \end{bmatrix}$

31. $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 1 & 0 & 3 \end{bmatrix}$

32. $A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & -1 & 3 \\ 5 & -4 & 3 \end{bmatrix}$

In Exercises 33 through 36, determine which of the linear systems have a solution by using Theorem 6.14.

33. $\begin{bmatrix} 1 & 2 & 5 & -2 \\ 2 & 3 & -2 & 4 \\ 5 & 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

34. $\begin{bmatrix} 1 & 2 & 5 & -2 \\ 2 & 3 & -2 & 4 \\ 5 & 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ -13 \\ 3 \end{bmatrix}$

35. $\begin{bmatrix} 1 & -2 & -3 & 4 \\ 4 & -1 & -5 & 6 \\ 2 & 3 & 1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$

36. $\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 5 & 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \\ 5 \end{bmatrix}$

In Exercises 37 through 40, compute the rank of the given bit matrix.

37. $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

38. $\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$

39. $\begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$

40. $\begin{bmatrix} 1 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix}$

Theoretical Exercises

T.1. Prove Corollary 6.2.

T.2. Prove Corollary 6.3.

T.3. Prove Corollary 6.4.

T.4. Let A be an $n \times n$ matrix. Show that the homogeneous system $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution if and only if the columns of A are linearly dependent.

T.5. Let A be an $n \times n$ matrix. Show that rank $A = n$ if and only if the columns of A are linearly independent.

T.6. Let A be an $n \times n$ matrix. Show that the rows of A are linearly independent if and only if the columns of A span R^n.

T.7. Let A be an $m \times n$ matrix. Show that the linear system $A\mathbf{x} = \mathbf{b}$ has a solution for every $m \times 1$ matrix $\mathbf{b}$ if and only if rank $A = m$.

T.8. Let A be an $m \times n$ matrix. Show that the columns of A are linearly independent if and only if the homogeneous system $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.

T.9. Let A be an $m \times n$ matrix. Show that the linear system $A\mathbf{x} = \mathbf{b}$ has at most one solution for every $m \times 1$ matrix $\mathbf{b}$ if and only if the associated homogeneous system $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.

T.10. Let A be an $m \times n$ matrix with $m \neq n$. Show that either the rows or the columns of A are linearly dependent.

T.11. Suppose that the linear system $A\mathbf{x} = \mathbf{b}$, where A is $m \times n$, is consistent (has a solution). Show that the solution is unique if and only if rank $A = n$.

T.12. Show that if A is an $m \times n$ matrix such that AA^T is nonsingular, then rank $A = m$.

MATLAB Exercises

Given a matrix A, the nonzero rows of **rref(A)** *form a basis for the row space of A and the nonzero rows of* **rref(A')** *transformed to columns give a basis for the column space of A.*

ML.1. Solve Exercises 1 through 4 using MATLAB.

To find a basis for the row space of A that consists of rows of A, we compute **rref(A')**. *The leading 1s point to the original rows of A that give us a basis for the row space. See Example 3.*

ML.2. Determine two bases for the row space of A that have no vectors in common.

(a) $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & 0 \\ 4 & 11 & 2 \\ 6 & 9 & 1 \end{bmatrix}$

(b) $A = \begin{bmatrix} 2 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 2 & 2 & 1 \\ 4 & 5 & 6 & 2 \\ 3 & 3 & 4 & 1 \end{bmatrix}$

ML.3. Repeat Exercise ML.2 for column spaces.

To compute the rank of a matrix A in MATLAB, *use the command* **rank(A)**.

ML.4. Compute the rank and nullity of each of the following matrices.

(a) $\begin{bmatrix} 3 & 2 & 1 \\ 1 & 2 & -1 \\ 2 & 1 & 3 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 2 & 1 & 2 & 1 \\ 2 & 1 & 0 & 0 & 2 \\ 1 & -1 & -1 & -2 & 1 \\ 3 & 0 & -1 & -2 & 3 \end{bmatrix}$

ML.5. Using only the rank command, determine which of the following linear systems is consistent.

(a) $\begin{bmatrix} 1 & 2 & 4 & -1 \\ 0 & 1 & 2 & 0 \\ 3 & 1 & 1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 21 \\ 8 \\ 16 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 2 & 1 \\ 1 & 1 & 0 \\ 2 & 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 2 \\ 2 & 0 \\ 2 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 3 \\ 2 \end{bmatrix}$

6.7 COORDINATES AND CHANGE OF BASIS

COORDINATES

If V is an n-dimensional vector space, we know that V has a basis S with n vectors in it; so far we have not paid much attention to the order of the vectors in S. However, in the discussion of this section we speak of an **ordered basis** $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ for V; thus $S_1 = \{\mathbf{v}_2, \mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is a different ordered basis for V.

If $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is an ordered basis for the n-dimensional vector space V, then every vector $\mathbf{v}$ in V can be uniquely expressed in the form

$$\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n,$$

where $c_1, c_2, \ldots, c_n$ are real numbers. We shall refer to

$$\left[\mathbf{v}\right]_S = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

as the **coordinate vector of v with respect to the ordered basis** S. The entries of $\left[\mathbf{v}\right]_S$ are called the **coordinates of v with respect to** S.

Observe that the coordinate vector $\left[\mathbf{v}\right]_S$ depends on the order in which the vectors in S are listed; a change in the order of this listing may change the coordinates of $\mathbf{v}$ with respect to S. All bases considered in this section are assumed to be ordered bases.